

Supplementary Material

Smaller 5-chromatic unit-distance graphs on two icosahedral spheres

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1 Overview

graph	sphere	parent	vertices	edges	properties
H_{231}	$S^2(r_1)$, $r_1 = \cos \frac{\pi}{10}$	G_{372}	231	938	$\chi = 5$, vertex-critical, no Moser spindle
H_{961}	$S^2(r_2)$, $r_2 = \cos \frac{3\pi}{10}$	G_{972}	961	4028	$\chi = 5$, vertex-critical, no Moser spindle

G_{372} and G_{972} are the graphs of Voronov, Neopryatnaya and Dergachev (Discrete Math. 345 (2022) 113106). H_{231} is the smallest graph found by our search; it is not claimed to be minimum. All files named below are in the repository accompanying the paper (<https://github.com/chandanshukla1989/5chromatic-spheres>); `verify_all.sh` runs every check.

2 Exact coordinates and number fields

For each sphere all coordinates lie in a number field K of degree 8 (Galois group of the Galois closure D_8), a tower of quadratic extensions $\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\sqrt{5}, b) \subset \mathbb{Q}(\sqrt{5}, b, g) = K$ with $b = \sqrt{\beta}$, $g = \sqrt{\gamma}$ (positive roots):

$$\begin{aligned} S^2(r_1) : \quad & \beta = \frac{7}{2} + \frac{5}{2}\sqrt{5} \approx 9.0902, \quad \gamma = \frac{15}{4} - \frac{7}{4}\sqrt{5} + \left(\frac{3}{4} - \frac{1}{4}\sqrt{5}\right)b \approx 0.41269, \\ S^2(r_2) : \quad & \beta = \frac{23}{2} + \frac{11}{2}\sqrt{5} \approx 23.798, \quad \gamma = \frac{5}{4} - \frac{3}{4}\sqrt{5} + \left(-\frac{1}{4} + \frac{1}{4}\sqrt{5}\right)b \approx 1.0804. \end{aligned}$$

Every coordinate is written as $\sum q_j e_j$ with rational q_j and $e_j \in \{1, \sqrt{5}, b, \sqrt{5}b, g, \sqrt{5}g, bg, \sqrt{5}bg\}$. The files `H231_exact_coordinates.m` and `H961_exact_coordinates.m` give all vertices in this form (Mathematica syntax, `Sqrt[...]`); the `.py` files give the same in Python syntax (`(x)**(1/2)`; load with `sympy` in rational mode). All vertices of H_{231} and the eleven vertices deleted from G_{972} are listed in Appendices A and B.

3 Parent graphs and index mapping

`construction/build_G372.py` and `build_G972.py` rebuild the parent graphs with 60-digit coordinates from the published minimal polynomials (and the closed form of the second coordinate of w_8). The *parent index* of a vertex is its position in the output: for G_{372} the icosahedron (0–11), then the orbits of v_1 (12–131), v_2 (132–251), v_3 (252–371); for G_{972} the orbits of $w_1, \dots, w_8$ (0–959, 120 each), then the great icosahedron (960–971). Within an orbit the order is that of a breadth-first enumeration of the group from its generators (as in the scripts). Column 2 of `H231_vertices.txt` and `H961_vertices.txt` gives the parent index; `verify_graph.py` checks that each vertex coincides with the parent vertex of that index (maximal difference $< 10^{-30}$, the precision of the files). The group Γ' used for G_{972} is generated, as in the authors' notebook, by g_1, g_2, Mg_3M, g_4 ; it equals $M\Gamma M$ (M : swap of y and z), which we checked by enumerating both groups.

4 Verification methodology

claim	method (program)
vertices on the sphere, edges of length 1	exact arithmetic in K with PARI/GP (<code>verification/pari/exact372.gp</code> , <code>exact972.gp</code>); independently, exact arithmetic on the radical files with an own implementation of the tower (<code>check_exact_radicals.py</code> ; perturbing a single coordinate was confirmed to be detected)
no missing unit edge	all non-adjacent pairs have $ p-q -1 \geq 0.0023$ (H_{231}), ≥ 0.00042 (H_{961}) with 30-digit coordinates (<code>verify_graph.py</code>)
no proper 4-colouring	DRAT proof checked by <code>drat-trim</code> ; cross-checks with CaDiCaL, Glucose, MiniSat and a binary encoding (<code>crosscheck_solvers.py</code>); SAT-free DSATUR search for H_{231} (<code>dsatur.c</code>)
proper 5-colouring	edge-by-edge check of <code>H*_5colouring.txt</code> (<code>verify_graph.py</code>)
vertex-critical	edge-by-edge check of one 4-colouring of $G-v$ per vertex v (<code>H*_criticality_certificates.txt</code> , <code>verify_graph.py</code>)
no Moser spindle	exhaustive search (<code>verify_graph.py</code>)

5 CNF encoding

For a graph with vertices $0, \dots, n-1$ the variable $4v+c+1$ means “vertex v has colour c ” ($c = 0, \dots, 3$). The files `H*_4col.cnf` contain, for every vertex, the clause $\bigvee_c x_{v,c}$ and the six clauses $\neg x_{v,c} \vee \neg x_{v,c'}$; for every edge uv and colour c the clause $\neg x_{u,c} \vee \neg x_{v,c}$; and two unit clauses giving colours 0 and 1 to the endpoints of the first edge (no loss of generality, by symmetry of colours). The formula is satisfiable if and only if the graph is 4-colourable.

6 DRAT verification

Proofs were produced with `kissat --no-binary H231_4col.cnf H231_4col.drat` (Kissat 4.0.4) and are stored compressed. To check:

```
xz -dk H231/H231_4col.drat.xz
drat-trim H231/H231_4col.cnf H231/H231_4col.drat -t 20000      # -> s VERIFIED
```

and likewise for H_{961} (about two minutes). Our `drat-trim` outputs are in `H*_drat-trim.log` and `results/drat-trim*.log`.

7 5-colourings, criticality, Moser spindles

`H*_5colouring.txt` lists *vertex colour* with colours 1–5. `H*_criticality_certificates.txt` has one line per vertex v : v string, where character x of the string is the colour (0–3) of vertex x in a proper 4-colouring of $G - v$ and $-$ for $x = v$. A Moser spindle is two rhombi (pairs of unit triangles sharing an edge) with a common apex whose tips are adjacent; `verify_graph.py` enumerates all of them (none found).

8 Software versions

Python 3.14.7; python-sat 1.9.dev15 (CaDiCaL 1.5.3, Glucose 4, MiniSat 2.2); mpmath 1.3.0; numpy 2.5.3; sympy 1.14.0; PARI/GP 2.19.0 with galddata; Kissat 4.0.4; drat-trim (commit 2e3b2dc, 2024-11-25); Apple clang 21.0.0; XZ Utils 5.8.4; macOS (Darwin, arm64).

9 Reproducing all checks

```
pip install -r requirements.txt
DRAT_TRIM=/path/to/drat-trim ./verify_all.sh          # about 6 minutes on a laptop
```

Individual steps and expected outputs are listed in `REPRODUCIBILITY.md`.

10 File manifest (SHA-256)

file	content
H231/H231.vertices.txt	vertex index, parent index in G_{372} , coordinates (30 digits)
0274f8a5682d48e57066312504f7d9bf35b59c4d838267e3e544eb93a19664ca	
H231/H231.edges.txt	edge list (0-based)
68ab49fb21ce19664e973dd3ea7b23cb2daa4e279582c6af8828b865469657b9	
H231/H231.exact_coordinates.m	exact coordinates, nested radicals, Mathematica syntax
08816dfb194f8aa2e4761a2fc74bacd5d332923810faa635460a610a7a7f47ff	
H231/H231.exact_coordinates.py	exact coordinates, Python syntax
c608218468272ded9a6469f9201d00d936c981674fcc3e955be1f459248fb615	
H231/H231.5colouring.txt	proper 5-colouring
f85d082a8f047ba21bc3e016fe1205bd1e1dcf872f110137d929d0ad0abae2a2	
H231/H231.criticality_certificates.txt	231 certificates: 4-colouring of $H_{231} - v$
d94b3378d3c22bdac64afca877c4d21a369fe7483b6e19dc69304733286e7679	
H231/H231.4col.cnf	CNF: H_{231} is 4-colourable
a50d0a4c2923460bdadb427f5dce60c3ff32ea54bae7d5718900bc2e843cd257	
H231/H231.4col.drat.xz	DRAT proof (xz)
f95c3f11c54ed80824923fb5aeeab50ea255452446c300b81ddb434ddcbdcf75	
H231/H231.drat-trim.log	drat-trim output
b0c3c26e8a3c8ec3a81c23ff85cf46092ba061b1f12cb7a94c8291e863051986	
H961/H961.vertices.txt	vertex index, parent index in G_{972} , coordinates (30 digits)
0aebc9420a5c6740059df93bfc8553519323c5d379401d710ff1791359f487d5	
H961/H961.edges.txt	edge list (0-based)
66df9964b7a216380ea53f808cd35c49424b2e8d5092580de79bf8cd1a2344b2	
H961/H961.exact_coordinates.m	exact coordinates, Mathematica syntax
35508ad690513bbe3274eac3763e62b60427374220449039855c6e825f54e11f	
H961/H961.exact_coordinates.py	exact coordinates, Python syntax
6fbbab8869cf552fdc48bccc7cd7043309f53bfa4ae431b2b17d980b5b3dd81e4	
H961/H961.5colouring.txt	proper 5-colouring
b523d8b71897994dc8b677ef8b8fbd98a921109e9d87796bc5c0299d21b22447	
H961/H961.criticality_certificates.txt	961 certificates: 4-colouring of $H_{961} - v$
6431a18ab0d3ee41deac0121cbfbc507dd8ec7f3d27e5e2f2e1b0da97b08d18e	
H961/H961.4col.cnf	CNF: H_{961} is 4-colourable
cd65022ae9349a07161a8135d037156a78590c91d4d3e3039f0f1bcd3566c8d0	

H961/H961.4col.drat.xz	DRAT proof (xz)
01dfb240bbd1025ada04a435c001a3114d2db47acf8de605e7bc0a2151b20966	
H961/H961.drat-trim.log	drat-trim output
d762619404a1fc33d5adc49ad22f29c45266d547cb3d29afa2ca5adaf2ac525d	
construction/build_G372.py	rebuilds G_{372} (60 digits)
a9ae2983a7ab7593c03dabb88f19938a14d9850209cf6ba6f93e5307b4736aa0	
construction/build_G972.py	rebuilds G_{972} (60 digits)
9b9d114caaacc9457cbbadea89fe8bff5a222a0ebc809a1476f2a5dfd956253e	
verification/verify_graph.py	counts, parent indices, edges, colourings, certificates, spin-dles
9241418d2a34aeab205887ade9a15028b1b3704c5bde0e8f2c278662bcb1b649	
verification/check_exact_radicals.py	exact check of the radical coordinates (no PARI/GP)
90e930b66dcf424a77cbea2018fb95d86bfe8acc85720c166fb8274ac7903605	
verification/crosscheck_solvers.py	three SAT solvers, two encodings
5b2f5e075b7a0ae2b178a86f52e3ea64638f3b0b8cc39e4e1fdf017186ccce14	
verification/dsatur.c	SAT-free exhaustive 4-colouring search
f91c8f7ffcdceb65f48397f44e11894f161cc550d2182c377c3a207cd30b0eac	
verification/make_criticality_certificates.py	generates the criticality certificates
74449286935b38ed55b2af956a517502410c0f9667bd740c71f1c65826b24078	
verification/pari/exact372.gp	exact verification of H_{231} in a degree-8 field
60dcbb3168f852bc4705d9a2c0664e71ae0f37bfea0e6c3b7ee6fc0b61ac8d4d	
verification/pari/exact972.gp	exact verification of H_{961} in a degree-8 field
bf749634bc5f26f633c93eee76e5266d499b8feb9cbe688d2c4b176c6e957462	
verify_all.sh	runs every check
5cb79565beb9d2cc148e1f61349cf95ce7e5be896ae4ac3e0d981bab6c7a3cbc	

A All vertices of H_{231} in radicals

Numbering as in Appendix A of the paper (vertex i here is index $i - 1$ in the edge list); parent index in G_{372} , orbit, and colour in the proper 5-colouring. b, g as in Section 2 for $S^2(r_1)$.

- 1 (parent 0, icosahedron, colour 1): $x = \frac{1}{4} + \frac{1}{4}\sqrt{5}, \quad y = \frac{1}{2}, \quad z = 0$
- 2 (parent 1, icosahedron, colour 5): $x = -\frac{1}{4} - \frac{1}{4}\sqrt{5}, \quad y = -\frac{1}{2}, \quad z = 0$
- 3 (parent 2, icosahedron, colour 4): $x = 0, \quad y = \frac{1}{4} + \frac{1}{4}\sqrt{5}, \quad z = \frac{1}{2}$
- 4 (parent 3, icosahedron, colour 3): $x = \frac{1}{4} + \frac{1}{4}\sqrt{5}, \quad y = -\frac{1}{2}, \quad z = 0$
- 5 (parent 4, icosahedron, colour 1): $x = 0, \quad y = -\frac{1}{4} - \frac{1}{4}\sqrt{5}, \quad z = -\frac{1}{2}$
- 6 (parent 5, icosahedron, colour 1): $x = -\frac{1}{4} - \frac{1}{4}\sqrt{5}, \quad y = \frac{1}{2}, \quad z = 0$
- 7 (parent 6, icosahedron, colour 4): $x = 0, \quad y = -\frac{1}{4} - \frac{1}{4}\sqrt{5}, \quad z = \frac{1}{2}$
- 8 (parent 7, icosahedron, colour 2): $x = \frac{1}{2}, \quad y = 0, \quad z = \frac{1}{4} + \frac{1}{4}\sqrt{5}$
- 9 (parent 8, icosahedron, colour 3): $x = -\frac{1}{2}, \quad y = 0, \quad z = \frac{1}{4} + \frac{1}{4}\sqrt{5}$
- 10 (parent 9, icosahedron, colour 3): $x = 0, \quad y = \frac{1}{4} + \frac{1}{4}\sqrt{5}, \quad z = -\frac{1}{2}$
- 11 (parent 10, icosahedron, colour 4): $x = -\frac{1}{2}, \quad y = 0, \quad z = -\frac{1}{4} - \frac{1}{4}\sqrt{5}$
- 12 (parent 11, icosahedron, colour 2): $x = \frac{1}{2}, \quad y = 0, \quad z = -\frac{1}{4} - \frac{1}{4}\sqrt{5}$
- 13 (parent 12, v_1 -orbit, colour 2): $x = \frac{1}{4} - \frac{1}{8}b + \frac{1}{8}\sqrt{5}b, \quad y = -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b, \quad z = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g$
- 14 (parent 13, v_1 -orbit, colour 5): $x = -\frac{1}{4} + \frac{1}{8}b - \frac{1}{8}\sqrt{5}b, \quad y = \frac{1}{8} + \frac{1}{8}\sqrt{5} - \frac{1}{4}b, \quad z = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g$
- 15 (parent 14, v_1 -orbit, colour 2): $x = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g, \quad y = \frac{1}{4} - \frac{1}{8}b + \frac{1}{8}\sqrt{5}b, \quad z = -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b$
- 16 (parent 15, v_1 -orbit, colour 2): $x = \frac{5}{16} + \frac{1}{16}\sqrt{5} - \frac{1}{8}b + \frac{1}{4}g, \quad y = -\frac{1}{16} + \frac{1}{16}\sqrt{5} + \frac{1}{16}b + \frac{1}{16}\sqrt{5}b - \frac{1}{8}g - \frac{1}{8}\sqrt{5}g, \quad z = -\frac{1}{8} + \frac{5}{16}b - \frac{1}{16}\sqrt{5}b + \frac{3}{8}g + \frac{1}{8}\sqrt{5}g$
- 17 (parent 17, v_1 -orbit, colour 3): $x = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g, \quad y = -\frac{1}{4} + \frac{1}{8}b - \frac{1}{8}\sqrt{5}b, \quad z = \frac{1}{8} + \frac{1}{8}\sqrt{5} - \frac{1}{4}b$
- 18 (parent 20, v_1 -orbit, colour 3): $x = -\frac{1}{4}g - \frac{1}{4}\sqrt{5}g, \quad y = -\frac{1}{4} + \frac{1}{8}b - \frac{1}{8}\sqrt{5}b, \quad z = -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b$
- 19 (parent 21, v_1 -orbit, colour 2): $x = -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b, \quad y = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g, \quad z = \frac{1}{4} - \frac{1}{8}b + \frac{1}{8}\sqrt{5}b$
- 20 (parent 22, v_1 -orbit, colour 2): $x = -\frac{3}{16} - \frac{1}{16}\sqrt{5} - \frac{3}{16}b + \frac{1}{16}\sqrt{5}b + \frac{1}{8}g + \frac{1}{8}\sqrt{5}g, \quad y = \frac{1}{8}\sqrt{5} + \frac{1}{16}b - \frac{1}{16}\sqrt{5}b + \frac{3}{8}g + \frac{1}{8}\sqrt{5}g, \quad z = -\frac{1}{16} - \frac{1}{16}\sqrt{5} + \frac{1}{8}\sqrt{5}b + \frac{1}{4}g$
- 21 (parent 23, v_1 -orbit, colour 2): $x = \frac{1}{4}g + \frac{1}{4}\sqrt{5}g, \quad y = -\frac{1}{4} + \frac{1}{8}b - \frac{1}{8}\sqrt{5}b, \quad z = -\frac{1}{8} - \frac{1}{8}\sqrt{5} + \frac{1}{4}b$

[illegible]

[illegible]

$$\begin{aligned}
\mathbf{1:} \quad & x = \frac{125}{656} - \frac{85}{656}\sqrt{5} + \frac{5}{41}b - \frac{19}{328}\sqrt{5}b - \frac{7}{164}g + \frac{21}{328}\sqrt{5}g - \frac{13}{656}bg - \frac{1}{656}\sqrt{5}bg, \\
& y = -\frac{111}{656} - \frac{59}{656}\sqrt{5} - \frac{135}{656}b + \frac{59}{656}\sqrt{5}b + \frac{119}{656}g - \frac{35}{656}\sqrt{5}g + \frac{1}{164}bg - \frac{3}{328}\sqrt{5}bg, \\
& z = -\frac{10}{41} + \frac{79}{328}\sqrt{5} + \frac{37}{656}b - \frac{35}{656}\sqrt{5}b + \frac{67}{656}g + \frac{43}{656}\sqrt{5}g - \frac{67}{656}bg + \frac{39}{656}\sqrt{5}bg \\
\mathbf{2:} \quad & x = -\frac{87}{328} + \frac{7}{82}\sqrt{5} - \frac{113}{656}b + \frac{67}{656}\sqrt{5}b - \frac{17}{656}g + \frac{87}{656}\sqrt{5}g + \frac{17}{656}bg - \frac{5}{656}\sqrt{5}bg, \\
& y = \frac{159}{656} + \frac{69}{656}\sqrt{5} + \frac{23}{328}b - \frac{7}{164}\sqrt{5}b + \frac{3}{164}g - \frac{9}{328}\sqrt{5}g + \frac{29}{656}bg - \frac{23}{656}\sqrt{5}bg, \\
& z = \frac{257}{656} - \frac{119}{656}\sqrt{5} - \frac{93}{656}b + \frac{37}{656}\sqrt{5}b - \frac{51}{656}g + \frac{15}{656}\sqrt{5}g - \frac{9}{82}bg + \frac{13}{328}\sqrt{5}bg \\
\mathbf{3:} \quad & x = \frac{141}{656} + \frac{55}{656}\sqrt{5} - \frac{91}{328}b + \frac{17}{164}\sqrt{5}b - \frac{79}{328}g + \frac{2}{41}\sqrt{5}g - \frac{47}{656}bg + \frac{9}{656}\sqrt{5}bg, \\
& y = \frac{319}{656} - \frac{89}{656}\sqrt{5} + \frac{9}{656}b + \frac{7}{656}\sqrt{5}b + \frac{79}{656}g + \frac{25}{656}\sqrt{5}g - \frac{19}{328}bg + \frac{1}{41}\sqrt{5}bg, \\
& z = \frac{109}{328} - \frac{5}{82}\sqrt{5} - \frac{13}{656}b - \frac{1}{656}\sqrt{5}b - \frac{23}{656}g - \frac{27}{656}\sqrt{5}g - \frac{59}{656}bg + \frac{27}{656}\sqrt{5}bg \\
\mathbf{4:} \quad & x = -\frac{125}{656} + \frac{85}{656}\sqrt{5} - \frac{5}{41}b + \frac{19}{328}\sqrt{5}b - \frac{7}{164}g + \frac{21}{328}\sqrt{5}g - \frac{13}{656}bg - \frac{1}{656}\sqrt{5}bg, \\
& y = \frac{111}{656} + \frac{59}{656}\sqrt{5} + \frac{135}{656}b - \frac{59}{656}\sqrt{5}b + \frac{119}{656}g - \frac{35}{656}\sqrt{5}g + \frac{1}{164}bg - \frac{3}{328}\sqrt{5}bg, \\
& z = \frac{10}{41} - \frac{79}{328}\sqrt{5} - \frac{37}{656}b + \frac{35}{656}\sqrt{5}b + \frac{67}{656}g + \frac{43}{656}\sqrt{5}g - \frac{67}{656}bg + \frac{39}{656}\sqrt{5}bg \\
\mathbf{5:} \quad & x = -\frac{23}{41} + \frac{15}{164}\sqrt{5} - \frac{21}{328}b + \frac{11}{328}\sqrt{5}b - \frac{17}{164}g + \frac{5}{164}\sqrt{5}g + \frac{17}{164}bg - \frac{5}{164}\sqrt{5}bg, \\
& y = -\frac{133}{328} + \frac{15}{328}\sqrt{5} + \frac{51}{328}b - \frac{15}{328}\sqrt{5}b + \frac{65}{328}g + \frac{5}{328}\sqrt{5}g + \frac{17}{328}bg - \frac{5}{328}\sqrt{5}bg, \\
& z = \frac{11}{164} + \frac{1}{41}\sqrt{5} - \frac{63}{328}b + \frac{33}{328}\sqrt{5}b - \frac{5}{82}g + \frac{15}{164}\sqrt{5}g - \frac{21}{328}bg + \frac{11}{328}\sqrt{5}bg \\
\mathbf{6:} \quad & x = -\frac{257}{656} + \frac{119}{656}\sqrt{5} + \frac{93}{656}b - \frac{37}{656}\sqrt{5}b + \frac{51}{656}g - \frac{15}{656}\sqrt{5}g + \frac{9}{82}bg - \frac{13}{328}\sqrt{5}bg, \\
& y = \frac{87}{328} - \frac{7}{82}\sqrt{5} + \frac{113}{656}b - \frac{67}{656}\sqrt{5}b + \frac{17}{656}g - \frac{87}{656}\sqrt{5}g - \frac{17}{656}bg + \frac{5}{656}\sqrt{5}bg, \\
& z = -\frac{159}{656} - \frac{69}{656}\sqrt{5} - \frac{23}{328}b + \frac{7}{164}\sqrt{5}b - \frac{3}{164}g + \frac{9}{328}\sqrt{5}g - \frac{29}{656}bg + \frac{23}{656}\sqrt{5}bg \\
\mathbf{7:} \quad & x = \frac{11}{164} + \frac{1}{41}\sqrt{5} - \frac{63}{328}b + \frac{33}{328}\sqrt{5}b - \frac{5}{82}g + \frac{15}{164}\sqrt{5}g - \frac{21}{328}bg + \frac{11}{328}\sqrt{5}bg, \\
& y = -\frac{23}{41} + \frac{15}{164}\sqrt{5} - \frac{21}{328}b + \frac{11}{328}\sqrt{5}b - \frac{17}{164}g + \frac{5}{164}\sqrt{5}g + \frac{17}{164}bg - \frac{5}{164}\sqrt{5}bg, \\
& z = -\frac{133}{328} + \frac{15}{328}\sqrt{5} + \frac{51}{328}b - \frac{15}{328}\sqrt{5}b + \frac{65}{328}g + \frac{5}{328}\sqrt{5}g + \frac{17}{328}bg - \frac{5}{328}\sqrt{5}bg \\
\mathbf{8:} \quad & x = -\frac{29}{164} + \frac{87}{328}\sqrt{5} - \frac{89}{656}b + \frac{31}{656}\sqrt{5}b - \frac{107}{656}g + \frac{17}{656}\sqrt{5}g + \frac{25}{656}bg - \frac{17}{656}\sqrt{5}bg, \\
& y = -\frac{107}{656} + \frac{99}{656}\sqrt{5} + \frac{37}{164}b - \frac{29}{328}\sqrt{5}b + \frac{71}{328}g - \frac{1}{82}\sqrt{5}g + \frac{63}{656}bg - \frac{33}{656}\sqrt{5}bg, \\
& z = \frac{49}{656} + \frac{29}{656}\sqrt{5} + \frac{33}{656}b - \frac{29}{656}\sqrt{5}b - \frac{11}{656}g - \frac{45}{656}\sqrt{5}g - \frac{15}{328}bg + \frac{1}{164}\sqrt{5}bg \\
\mathbf{9:} \quad & x = -\frac{111}{656} - \frac{59}{656}\sqrt{5} - \frac{135}{656}b + \frac{59}{656}\sqrt{5}b - \frac{119}{656}g + \frac{35}{656}\sqrt{5}g - \frac{1}{164}bg + \frac{3}{328}\sqrt{5}bg, \\
& y = -\frac{10}{41} + \frac{79}{328}\sqrt{5} + \frac{37}{656}b - \frac{35}{656}\sqrt{5}b - \frac{67}{656}g - \frac{43}{656}\sqrt{5}g + \frac{67}{656}bg - \frac{39}{656}\sqrt{5}bg, \\
& z = \frac{125}{656} - \frac{85}{656}\sqrt{5} + \frac{5}{41}b - \frac{19}{328}\sqrt{5}b + \frac{7}{164}g - \frac{21}{328}\sqrt{5}g + \frac{13}{656}bg + \frac{1}{656}\sqrt{5}bg \\
\mathbf{10:} \quad & x = \frac{319}{656} - \frac{89}{656}\sqrt{5} + \frac{9}{656}b + \frac{7}{656}\sqrt{5}b + \frac{79}{656}g + \frac{25}{656}\sqrt{5}g - \frac{19}{328}bg + \frac{1}{41}\sqrt{5}bg, \\
& y = -\frac{109}{328} + \frac{5}{82}\sqrt{5} + \frac{13}{656}b + \frac{1}{656}\sqrt{5}b + \frac{23}{656}g + \frac{27}{656}\sqrt{5}g + \frac{59}{656}bg - \frac{27}{656}\sqrt{5}bg, \\
& z = -\frac{141}{656} - \frac{55}{656}\sqrt{5} + \frac{91}{328}b - \frac{17}{164}\sqrt{5}b + \frac{79}{328}g - \frac{2}{41}\sqrt{5}g + \frac{47}{656}bg - \frac{9}{656}\sqrt{5}bg \\
\mathbf{11:} \quad & x = \frac{107}{656} - \frac{99}{656}\sqrt{5} - \frac{37}{164}b + \frac{29}{328}\sqrt{5}b - \frac{71}{328}g + \frac{1}{82}\sqrt{5}g - \frac{63}{656}bg + \frac{33}{656}\sqrt{5}bg, \\
& y = \frac{49}{656} + \frac{29}{656}\sqrt{5} + \frac{33}{656}b - \frac{29}{656}\sqrt{5}b - \frac{11}{656}g - \frac{45}{656}\sqrt{5}g - \frac{15}{328}bg + \frac{1}{164}\sqrt{5}bg, \\
& z = \frac{29}{164} - \frac{87}{328}\sqrt{5} + \frac{89}{656}b - \frac{31}{656}\sqrt{5}b + \frac{107}{656}g - \frac{17}{656}\sqrt{5}g - \frac{25}{656}bg + \frac{17}{656}\sqrt{5}bg
\end{aligned}$$